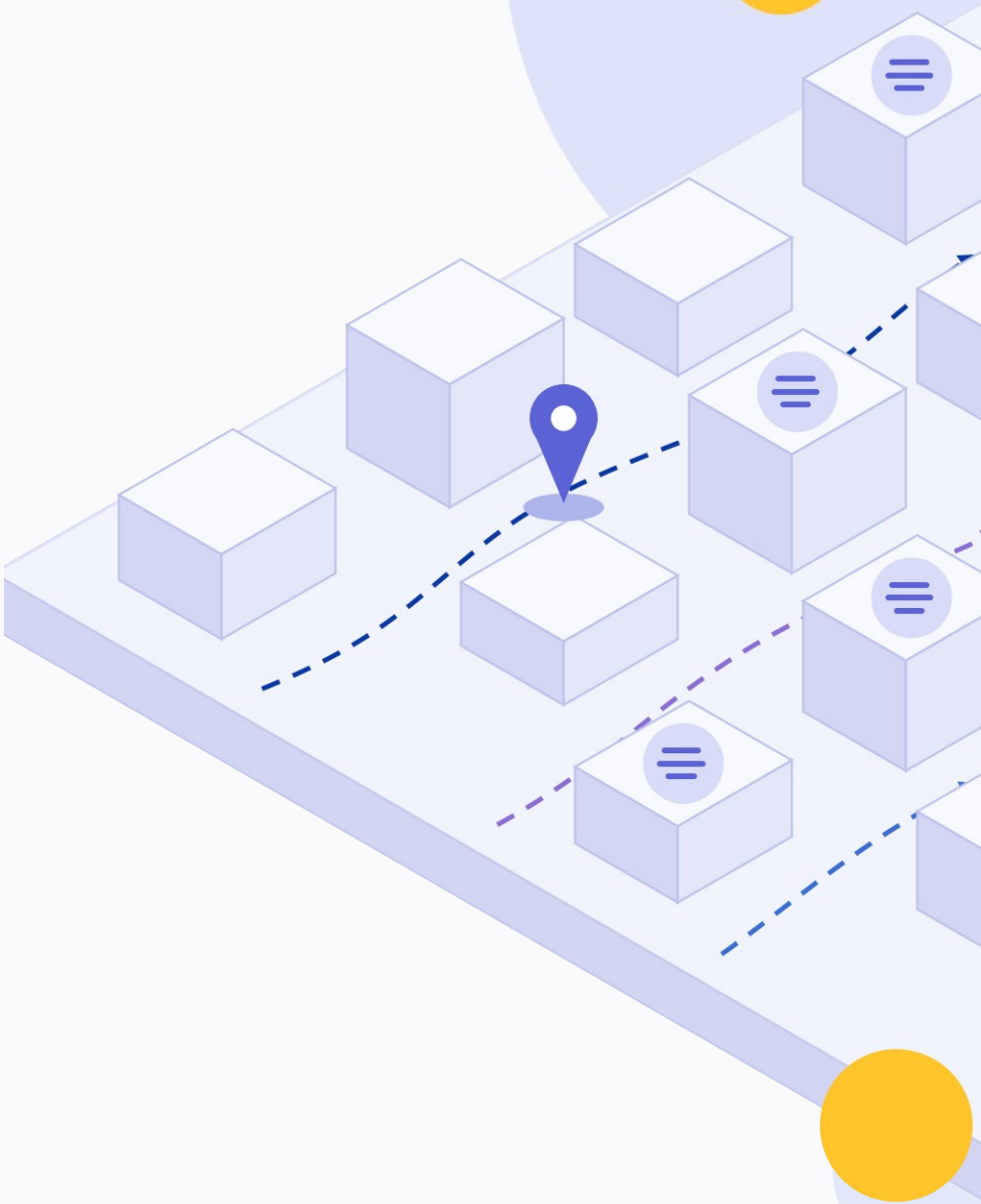
B.TECH PROJECT · TEAM T095 · SECTION B

Urban Microclimate Modelling via Neural Surrogates and Computational Fluid Dynamics



Team Members

Student Name	Enrollment Number
Aditya Mathur	2401010028
Vaidehi Sahu	2401010488
Om Kar Shukla	2401010307
Yachna	2401010509
Prateek Shukla	2401010342

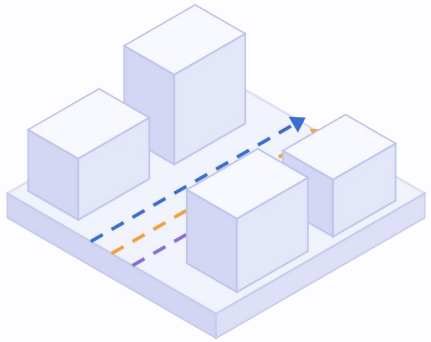
The Problem: Why Is Urban Wind Hard to Predict?

Background

Buildings **channel and block wind**, creating street-canyon jets, wake regions and local acceleration.

CFD can model these flows but needs **substantial compute time** and set-up.

Pedestrian-level wind affects **comfort, safety and pollutant dispersal**.



Why neural surrogates?

A trained network can approximate the **whole wind field in one forward pass** (speed not yet measured).

Spatial networks read the full layout image; **tabular models** see only summary features.

Goal: a **research-software pipeline** to compare both approaches fairly.

Mathematical problem formulation

Layout + inflow → 7-channel raster → predicted wind field:

$$\hat{U}(x, y) = f\theta(\, H, M, \varphi, U_in, V_in, X, Y \,)$$

Scalar baseline target: outdoor-masked area average of wind speed:

$$\bar{U} = \Sigma\, (1 - M) \cdot U \cdot \Delta A \;/\; \Sigma\, (1 - M) \cdot \Delta A$$

Project objectives

- **Validated data contract** and leakage-safe splits
- **26-D features** and **7-channel raster** inputs
- Compare **Random Forest, U-Net, dilated CNN**
- **Outdoor-masked evaluation** and integrity audit

Scope: 2D, time-averaged wind speed at **z = 1.75 m** under neutral, isothermal flow. No temperature, solar, buoyancy or 3D claims.

Data: Synthetic Corpus and Analytical Reference

No external UrbanTALES/CEDVAL data and no OpenFOAM output were used in the reported experiments.

24
Procedural layouts (4 named + 20 variants)

1
Canonical analytical reference

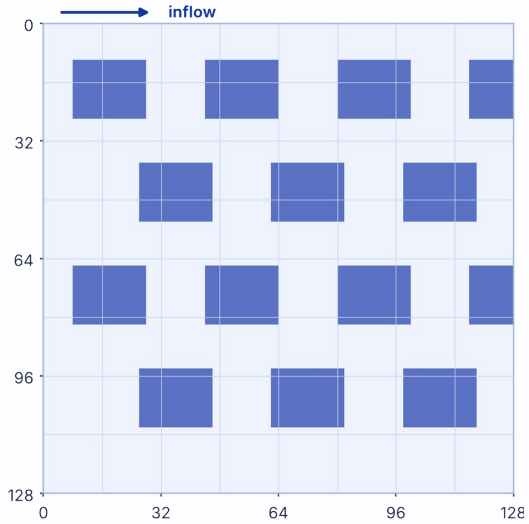
128 × 128
Spatial grid (16,384 cells)

14,080
Fluid cells in layout 042

Corpus characteristics

- Fields are **analytically synthesised**: stagnation, corner jets and a wake cavity (Castro & Robins-style).
- Target is **pedestrian wind speed at z = 1.75 m**; values are synthetic, not measured.
- Every layout passes the **data contract**: positive heights, finite values, matching shapes.
- Provenance flag **has_cfd_ground_truth = False** for all layouts (corrected in P0).

Layout grid schematic (128 × 128)



Building footprint (mask = 1)

Outdoor fluid cell (evaluated)

Cell size and domain size of corpus layouts: **Not recorded**.

Illustrative schematic only. It is not layout 042 data and not a CFD result.

Provenance tiers: what the project actually used

Synthetic fixtures	IN USE
Analytical reference	IN USE
UrbanTALES (LES)	NOT OBTAINED
OpenFOAM (RANS)	NOT RUN

Integrity Audit and Leakage-Safe Evaluation

Evaluation pipeline

Data validation contract checks → **Preprocessing** train-fitted scaling → **Features / raster** 26-D + 7 channels → **Layout-level split** whole layouts → **Model inference** RF · U-Net · CNN → **Outdoor mask** fluid cells only → **Metrics** MAE · RMSE · peak

Scaling statistics are fitted on training layouts only; splits are by whole layout or family, never by pixel.

Integrity audit: before → after

Issue identified	Action taken
Analytical fields labelled **OpenFOAM**	Provenance corrected: **has_cfd_ground_truth = False**
Neural "predictions" were **perturbed targets**	**Genuine PyTorch forward pass** with outdoor mask
Sleep loop, random MAE, hard-coded KPIs	**Fabricated metrics removed**; real background training
API and UI keys mismatched; metrics non-nullable	**Schemas aligned**; nullable performance fields
No view of data, checkpoints or CFD status	**/api/inventory/** endpoints added

Leakage-safe evaluation

- Split by **whole layout or family**, never random pixels, to avoid spatial leakage.
- Fit normalisation on **training layouts only**; invert exactly at test time.
- Score only **outdoor fluid cells** so building interiors cannot flatter metrics.

Outcome: Fabricated outputs were removed. The audit's original R^2 of 0.9881 came from a perturbed target and is **not** a model result.

Research Context and Gaps

Background sources (cited in report)

Source	Topic	Role in this project
Castro & Robins (1977)	Flow around a surface-mounted cube	Basis of analytical benchmark: stagnation, jets, wake
Macdonald et al. (1998)	Roughness of obstacle arrays	z0 and zd features for the RF baseline
Breiman (2001)	Random forests	Tabular baseline model
Ronneberger et al. (2015)	U-Net	Encoder-decoder with skip connections
Yu & Koltun (2016)	Dilated convolutions	Wide receptive field, no downsampling
Nazarian et al. (2025)	UrbanTALES LES dataset	Candidate data source; **not obtained**

Research context only: no performance figures from these works are compared in this presentation.

Research gaps

1. Scores can be **inflated** by pixel leakage and building-interior cells.
2. **Synthetic** or mislabelled data can masquerade as CFD output.
3. Scalar and spatial models are rarely compared under **one protocol**.
4. **Real LES/CFD data** and trained surrogates are still needed.

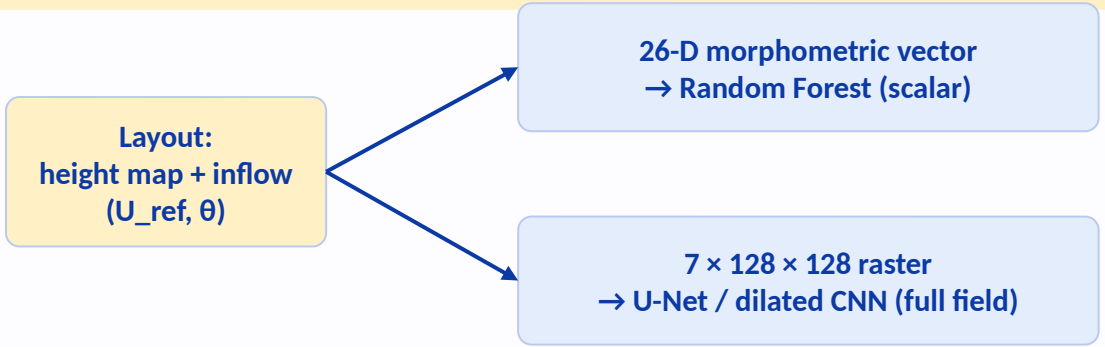
Project objectives

- Build a **validated data contract** and splits
- Create **features** and **raster inputs**
- Compare **RF, U-Net** and **dilated CNN**
- **Evaluate honestly** and audit claims

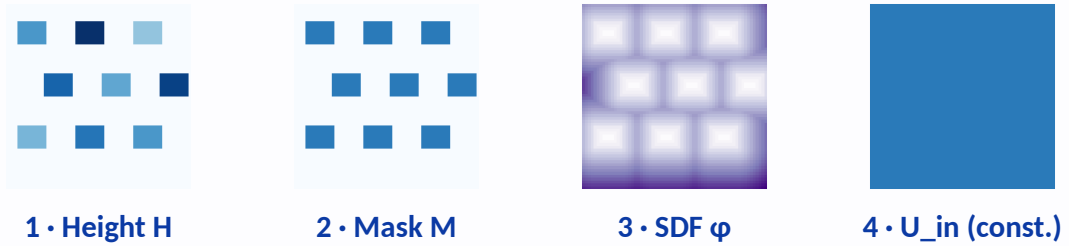
Central question: How do a scalar baseline and spatial surrogates compare under a **leakage-free, outdoor-masked protocol** once real data and trained models exist?

Input Representations: Features vs Raster

Input construction pipeline



Seven raster channels (illustrative toy layout)



Illustrative toy layout, not project data.

Two representations compared

Aspect	26-D vector	7-channel raster
Used by	Random Forest	U-Net and dilated CNN
Content	Summary statistics: λp, λf, height moments, z0, zd	Full layout, distance field and inflow
Predicts	One scalar per layout	Whole 128 × 128 field
Spatial structure	Not represented	Preserved

26-D feature groups

- Plan area fraction λp
- Frontal area index λf(θ)
- Height moments
- Centroids and footprints
- Domain and inflow descriptors
- Roughness z0 and zd

Design hypothesis: the RF cannot see where buildings are, the CNNs can. **No ablation has been run**, so no benefit of any channel is claimed.

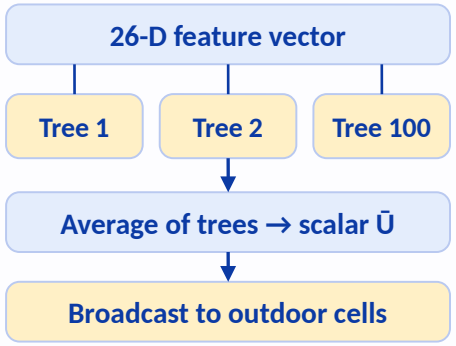
Model Architectures

1 Random Forest (RF)

100 trees, maximum depth **12**

Input: **26-D geometric vector**

Output: **one area-averaged speed**, broadcast over outdoor cells



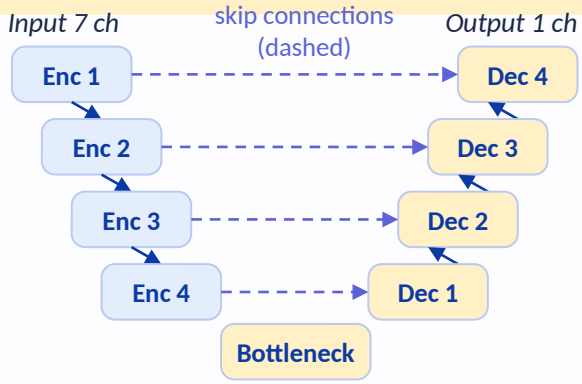
2 Compact residual U-Net

4-level encoder–decoder

Residual double-conv blocks

GroupNorm + **LeakyReLU**

Skip connections; mask sets U = 0 inside buildings



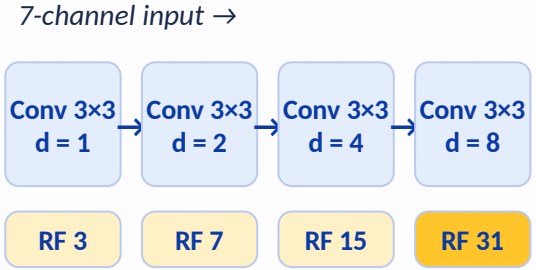
3 Compact dilated CNN

Four 3 × 3 convolutions

Dilation rates **1, 2, 4, 8**

Receptive field **31 × 31**

No downsampling: full resolution kept



Receptive field grows 3 → 7 → 15 → 31 cells

4 Purpose and trade-offs

Model	Purpose	Trade-off
Random Forest	Cheap tabular baseline	Scalar only; spatially blind
U-Net	Multi-scale context with skip detail	Needs training; parameters not recorded
Dilated CNN	Wide context at full resolution	No multi-scale encoder; parameters not recorded

Key takeaway: All three models share **one outdoor-masked protocol**. The neural checkpoints in the reported comparison are **untrained**.

Reported Results on Synthetic Layout 042

Reported errors: layout 042 (synthetic, analytical reference)

Model	MAE [m/s]	RMSE [m/s]	Peak [m/s]	Weights status
Random Forest	1.127	1.492	4.979	Fitted on synthetic corpus; held-out status **unverified**
Compact U-Net	2.608	3.084	5.191	**Untrained** (is_trained = False)
Compact dilated CNN	2.909	3.376	5.684	**Untrained** (is_trained = False)

Evaluation metrics

MAE
Mean $|\hat{y} - y|$ over outdoor fluid cells

RMSE
Root of mean squared error; penalises large misses

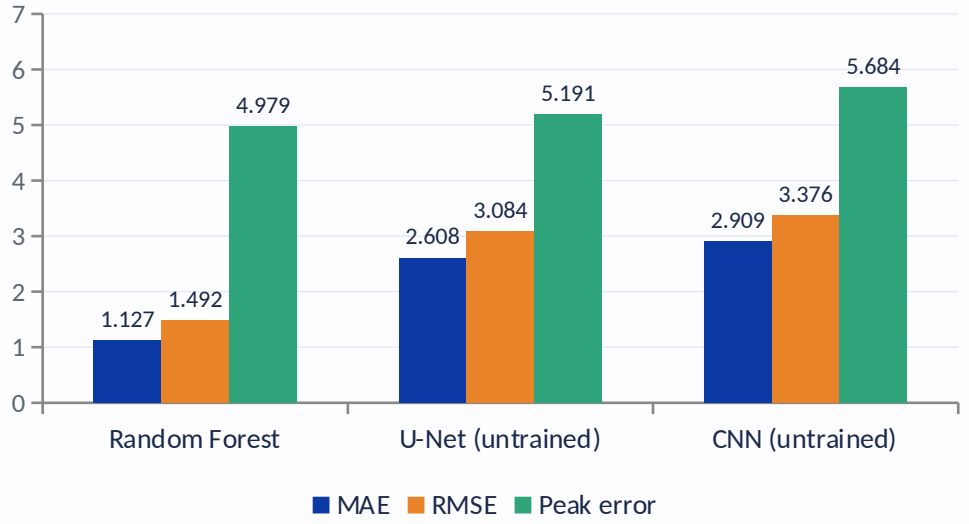
Peak error
Maximum $|\hat{y} - y|$ anywhere outdoors

NRMSE
Not reported for this comparison

R^2
Not reported for this comparison

Fluid cells
14,080 of 16,384 grid cells

MAE, RMSE and peak error [m/s]



Read with care

- Neural checkpoints are **untrained**.
- RF held-out status on layout 042 is **unverified**; it predicts one scalar, broadcast.
- Synthetic geometry and analytical reference: **not a fair model ranking**.

Key interpretation: These numbers verify the **evaluation pipeline**, not model skill. No model is ranked and no speed-up is claimed.

Scientific Interpretation and Next Steps

Implemented and reported tested

Data contract, rasteriser and **26 features**

Layout/family splits and train-only normalisers

RF, U-Net, dilated CNN, trainer and **masked metrics**

FastAPI and inventory endpoints; **207 tests reported** (breakdown unresolved)

What the current results show

The **evaluation pipeline** runs end to end

A **scalar baseline is spatially blind**: peak error 4.979 m/s

Provenance flags and **outdoor masking** behave as designed

Untrained networks give **arbitrary** outputs

Not yet scientifically established

Neural vs Random Forest **accuracy**

Any **real-world** accuracy or generalisation

RF **held-out** score on layout 042

CFD speed-up; no UrbanTALES/CEDVAL data, no OpenFOAM run

Needed for genuine validation

Obtain **verified LES/CFD data**; complete OpenFOAM runs

Train models; keep checkpoints with split hashes

Leakage-safe **family holdout** and ablations (SDF, U-Net vs CNN)

Benchmark inference against solver runtime

Position: a **verified evaluation framework**, not yet a validated surrogate. The scientific comparison remains open.

TEAM T095 · SECTION B

Thank You

Urban Microclimate Modelling via Neural Surrogates and Computational Fluid Dynamics

